Code Artifacts — Vehicle Washing Management System

Cover Page

Table 1

Document identification data

Field	Value
Project	Vehicle Washing Management System (Vehicle Washing / LavaRápido)
Deliverable	Code artifacts (section 2.5.4)
Program	Software Analysis and Development Technologist (ADSO)
Group (ficha)	3145556
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Training center	Industria
Date	October 21, 2026

The system's code lives in 10 GitHub repositories: the web application, the mobile application, and 8 backend repositories. Each one has a GitHub Actions workflow (`.github/workflows/ci.yml`) that builds it and runs its tests on every change (in `lavarapido-infra` it validates the deploy scripts, the Docker Compose files and the end-to-end test syntax); `lavarapido-infra` also has `deploy.yml`, a workflow started by hand to update the server. This document inventories the two clients and explains how to build and run them; the backend is described only as far as the clients need it. Source code is not reproduced.

1. Repository Inventory

Table 2

System repositories (`develop` branch, as of October 21, 2026)

Artifact	Repository	Technology	Commits	Last commit
Web application	Front-end-proyecto-web	Angular 21.2.25, TypeScript 5.9	474	2026-10-21
Mobile application	LavadoVehicular-Movil	React Native 0.86.3, Expo ~57, TypeScript 6.0	236	2026-10-21
Infrastructure	lavarapido-infra	Docker Compose	86	2026-10-21
API Gateway	lavarapido-api-gateway	Spring Boot 4.0.8, Java 21	21	2026-10-21

Artifact	Repository	Technology	Commits	Last commit
security-service	lavarapido-security-service	Spring Boot 4.0.8, Java 21	63	2026-10-21
customer-service	lavarapido-customer-service	Spring Boot 4.0.8, Java 21	22	2026-10-19
booking-service	lavarapido-booking-service	Spring Boot 4.0.8, Java 21	61	2026-10-21
operations-service	lavarapido-operation-service	Spring Boot 4.0.8, Java 21	50	2026-10-21
payment-service	lavarapido-payment-service	ASP.NET Core 8	49	2026-10-21
notification-service	lavarapido-notification-service	Spring Boot 4.0.8, Java 21	42	2026-10-21

Table 3

Client size

Artifact	Source files (`.ts`, `.tsx`, `.html`, `.scss`)	Lines	Screens
Web (`Web/src`)	449	48,202	35 routes
Mobile (`LavadoVehicular-Movil/src`)	296	35,540	31 screens

> *Note.* Count of October 21, 2026, with `git ls-files` and `wc -l` over `src`; includes tests and comments.

2. Web Application

The web client is an Angular 21 SPA (Angular, n.d.) with standalone components, one lazy-loaded module per role, and translations into four languages (es, en, fr, pt). Every call goes to `/api/v1`, which the development server proxies to the gateway at `localhost:8080`.

2.1 Structure

Table 4

Folders in `Web/src/app`

Folder	Contents
`core/constants`	`API_BASE_URL = '/api/v1'` (`core/constants/api.ts:4`)
`core/guards`	`authGuard` (session and role) and `guestGuard` (hides login from users who already have a session)

Folder	Contents
core/interceptors	authInterceptor: adds the JWT and ends the session on a 401 while the session is still open (it ignores a late 401 after logging out)
core/services	17 HTTP and state services: auth, booking-api, vehicles, operations-api, payments-api, promotions-api, loyalty-api, notifications, preferences, role-permissions-api, user-admin, user-session, inspection-api, inspection-photos, local-storage, notification-watcher, api; auth also follows sign-in and sign-out in the other tabs of the same browser
core/models, core/utils	DTO interfaces and utilities (for example, API error translation and image-compress.ts, which shrinks the receipt image to at most 2 MB before uploading it)
features/landing	Public page: navbar, hero, about us, features, services, download, contact, frequently asked questions, and footer
features/auth	Login, sign-up, and three-step password recovery
features/client	8 client pages + viewmodels/client.viewmodel.ts (layout)
features/operator	8 operator pages
features/admin	13 administrator pages + 5 stores (reservations, payments, operators, schedule, business)
features/not-found	Address not found page (route **)
features/public-report	Read-only inspection report (/report/:token)
shared/components, shared/dialogs, shared/pipes	12 reusable components and 20 dialogs (confirm, assign operator, rate, review payment, export…)
assets/i18n	es.json, en.json, fr.json, pt.json

2.2 Main Dependencies

Table 5

Web libraries (Web/package.json)

Library	Version	Purpose
@angular/* (core, router, forms, material, cdk)	21.2.25 (Material and CDK 21.2.14)	Framework and UI components
@ngx-translate/core, http-loader	^17.0.0	Translations
chart.js	^4.5.1	Dashboard and report charts
jspdf, jspdf-autotable	^4.2.1, ^5.0.8	Export reports to PDF
xlsx, file-saver	0.20.3 (SheetJS CDN), ^2.0.5	Export to Excel and download files
iconify-icon	^3.0.2	Icons
vitest, jsdom (development)	^4.1.7, ^28.0.0	Unit tests with @angular/build:unit-test

2.3 Build and Run

Requirements: Node.js 24 or a version compatible with Angular 21, and npm 11 (`packageManager: npm@11.6.2`). Screens load data only when the backend is running (section 4).

```
cd Web
npm install          # install dependencies
npm start            # ng serve at http://localhost:4200, proxying /api ->
http://localhost:8080
npm run build        # production build in dist/web/browser
npm test             # unit tests (Vitest)
```

The proxy lives in `Web/proxy.conf.json` and is enabled in `angular.json` (`proxyConfig`). The production build has size budgets: it fails if the initial bundle exceeds 1 MB or a component style exceeds 16 kB (`angular.json`, `budgets`). In production, `dist/web/browser` can be served by any static server that forwards `/api` to the gateway.

According to SRS v7 (RNF-008), the web client must work on current versions of Chrome, Firefox, and Edge and be usable from 360 px wide.

3. Mobile Application

The mobile app is built with Expo 57 on React Native 0.86 (Expo, n.d.; React Native, n.d.) and targets Android 8.0 or later (RNF-008 in SRS v6). The web build (`react-native-web`) is used only for development and testing. It talks only to the gateway and discovers the IP of the computer running Expo by itself, so no setup is needed on a local network.

3.1 Structure

Table 6

Folders in `LavadoVehicular-Movil/src`

Folder	Contents
`app/config`	i18next with `es`, `en`, `fr`, `pt` (`app/config/locales`)
`app/providers`	`AppProviders`, sync of the account preferences and the push token, and `AppUpdatesCheck` (if the server published a newer APK, it must be installed before going on)
`app/theme`	Theme context, colors, and typography
`core/api`	`config.ts` (gateway URL and `WEB_URL` for public links), `httpClient.ts` (fetch with JWT, JSON or *multipart*, and a 15 s timeout or the one each call asks for), `apiError.ts`
`core/navigation`	`RootNavigator` (screens per role) and route types

Folder	Contents
core/services	auth, booking, vehicles, operations, payments (with receiptImage.ts, which compresses the receipt to at most 2 MB), notifications, users, appUpdates (reads /app/version.json)
core/storage	Session in expo-secure-store (AsyncStorage on web)
modules/auth, client, operator, admin	Screens, components, models, hooks, and view models for each role
shared	Per-role layouts, shared views (profile, notifications, settings), validators, components, and utilities

3.2 Main Dependencies

Table 7

Mobile libraries (LavadoVehicular-Movil/package.json)

Library	Version	Purpose
expo	~57.0.30	Build and runtime platform
react, react-native	19.2.3, 0.86.3	User interface
@react-navigation/native, native-stack	^7.3.13, ^7.18.5	Navigation
expo-secure-store, @react-native-async-storage/async-storage	~57.0.4, 2.2.0	Session storage
i18next, react-i18next	^26.4.1, ^17.0.13	Translations
expo-print, expo-sharing, expo-file-system, expo-document-picker	~57.0.x	Export reports and attach receipts
expo-image-picker, expo-image-manipulator	~57.0.20, ~57.0.21	Take or pick the inspection report photos and shrink them before uploading
expo-notifications, expo-updates	~57.0.23, ~57.0.26	Push notifications and over-the-air (OTA) updates
react-native-svg, react-native-reanimated	15.15.4, 4.5.1	Charts and animations

3.3 Configuration

The backend URL is chosen in this order (src/core/api/config.ts):

1. EXPO_PUBLIC_API_URL in .env.local, if defined.
2. The IP of the computer running Expo (the same one the phone used to open the app).
3. Per platform: http://10.0.2.2:8080/api/v1 on the Android emulator and http://localhost:8080/api/v1 on the web build.

The `.env.example` template explains the optional variables; `.env.local` is not committed. The phone and the computer must be on the same Wi-Fi network, and the firewall must allow port 8080.

3.4 Build and Run

```
cd LavadoVehicular-Movil
npm install
npm start                # expo start: scan the QR code with Expo Go
npm run android          # open on an Android emulator or device
npm run web              # web version at http://localhost:8081
npx expo start --clear   # restart and clear the cache after changing .env.local
```

For development the app runs through Expo Go. The APK is built with EAS (`eas.json`, `preview` and `production` profiles), and the `npm run publish:apk` and `npm run publish:update` scripts publish a new APK on the server or an over-the-air update (see the Deployment Document, folder 07).

4. Backend Required by the Clients

The web and mobile clients only need the gateway at `localhost:8080`; behind it, the six services, SQL Server, and (if events are enabled) RabbitMQ must be running. Everything starts from `Backend/lavarapido-infra` with Docker.

```
cd Backend/lavarapido-infra
cp .env.example .env                          # fill in secrets; .env is never committed
docker compose up -d                          # only SQL Server, Mailpit, and RabbitMQ
(services run from the IDE)
docker compose --profile app up -d --build    # the whole backend in containers
```

Table 8

Variables in `.env.example` (no values)

Group	Variables
Database	`DB_PASSWORD` (required), `DB_NAME`, `DB_USERNAME`, `DB_URL`
Security	`JWT_SECRET` (required), `INTERNAL_API_KEY` (calls between services and automatic assignment), `CORS_ALLOWED_ORIGINS`
Email	`MAIL_ENABLED`, `MAIL_HOST`, `MAIL_PORT`, `MAIL_USERNAME`, `MAIL_PASSWORD`, `MAIL_FROM`
Messaging	`MESSAGING_ENABLED`, `RABBITMQ_HOST`, `RABBITMQ_PORT`, `RABBITMQ_USERNAME`, `RABBITMQ_PASSWORD`; `AUTO_ASSIGN_ENABLED=false` (optional) turns off only the automatic assignment

Group	Variables
Push	`PUSH_ENABLED` (and `EXPO_ACCESS_TOKEN` in the compose file)

> *Note.* The variables of October 21, 2026 are not in `.env.example` or in the compose file because they have defaults: `SESSION_CHECK_ENABLED`, `SESSION_CHECK_CACHE_TTL`, and `SESSION_CHECK_TIMEOUT` (gateway session check: on, 30 s, and 3 s); `BOOKING_EXPIRY_CHECK_ENABLED` and `BOOKING_EXPIRY_CHECK_INTERVAL` (cancellation of scheduled bookings that expired without confirmation, in booking-service: on and every 5 minutes); `BOOKING_EXPIRY_GRACE_MINUTES` (60-minute grace after the end of the booking, used by booking-service and operations-service); and `FOLLOW_UP_ENABLED` and `FOLLOW_UP_INTERVAL` (follow-up of the bookings that ended and nobody closed, in operations-service: on and every 5 minutes; it needs `INTERNAL_API_KEY`). They only change when the service runs from the IDE or when they are added to the service `environment` in the compose file.

To run a Java service from the console, use its Maven Wrapper (`./mvnw spring-boot:run` inside the service folder) with Java 21 (Spring, n.d.). payment-service runs with `dotnet run --project src/PaymentService.Api` and requires the .NET 8 SDK (Microsoft, n.d.). When starting with `--profile app`, the folders must be named after their repositories (`lavarapido-security-service`, `lavarapido-operation-service`), because that is how the compose file references them.

Table 9

Ports to check that everything is up

Service	Health check URL
API Gateway	`http://localhost:8080/actuator/health`
Java services	`http://localhost:300X/actuator/health` (X = 1, 2, 3, 4, 6) and Swagger at `/swagger-ui.html`
payment-service	`http://localhost:3005/actuator/health` and `/health` (both check SQL Server and answer 503 when it does not respond); Swagger at `/swagger` only in the Development environment (`Program.cs:53-57`)
RabbitMQ	Dashboard at `http://localhost:15672`
Mailpit	Inbox at `http://localhost:8025`

With everything up, `lavarapido-infra/e2e` has three kinds of Node 22 scripts: the 19 end-to-end tests (`node --test "e2e/*.test.js"`: cross-service booking flow, full flow of the three roles, inspection report and access rules), the response time measurement (`node e2e/performance.js`) and the check of the web at 360 px wide (`node e2e/width-360.js`, after `npm install --prefix e2e`). They need `MESSAGING_ENABLED=true` and `INTERNAL_API_KEY` in the `.env`.

References

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